Duke Robotics Club: Sensor Printed Circuit Board

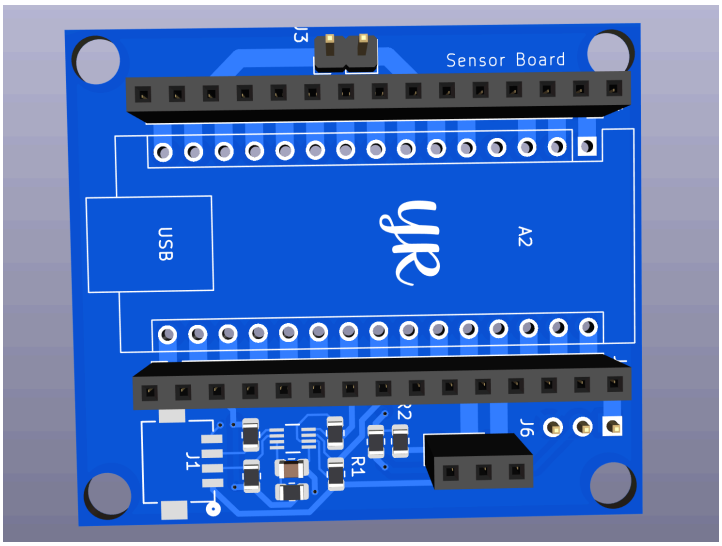
Date Started: February 19th 2026

Executive Summary:

The Sensor PCB is a custom board designed for Duke Robotics' RoboSub autonomous underwater robot, Crush. It combines scattered sensor wiring previously used into a single, standardized module that mounts directly onto the power distribution board. The board is built around an Arduino Nano v3 and integrates four sensing functions:

1. Battery voltage monitoring through a resistor divider feeding the ADC.
2. Pressure sensing via a Blue Robotics Bar30 connected over I2C through a level shifter.
3. Temperature and humidity reading through a DHT11 breakout module.
4. Gyro trigger interface exposing two Arduino timer pins through a screw terminal.

Every Arduino pin is broken out through female sockets on both sides of the board for debugging and future expansion. The board measures 41.5 × 48 mm, is 2-layer FR-4 (TG135) with 1.6 mm thickness, 1 oz outer copper, blue solder mask, and white silkscreen, finished with lead-free HASL. It was designed in KiCad and fabricated through JLCPCB. The bare boards weigh approximately 32 g each.



Design Objectives:

The PCB replaces scattered wiring and separate integrated circuits previously used across Oogway and Brush with a single board that slots into a stacked PCB assembly. The existing setup worked but was inconsistent between the two robots, hard to debug underwater, and difficult for anyone outside the person who built it to maintain. Moving to a PCB stack cuts the overall electrical stack size by roughly 20%, makes the system fully modular with every board

being plug and play, and improves reliability by eliminating loose jumper wires that can shift or disconnect under vibration and handling.

The board had to meet four requirements:

- To centralize every onboard sensor into one Arduino-driven interface: battery voltage, depth (pressure), temperature and humidity, and a gyro trigger line.
- To bridge the voltage gap between the Arduino's 5V I2C bus and the Bar30 pressure sensor's 3.3V logic, without requiring extra wiring or external level shifters.
- To physically mount onto the power board rather than exist as a loose module since space inside the sub is tight, and loose boards shift during competition handling.
- To be cheap enough to replicate: both subs run the same electrical stack, so one design has to cover both, and the cost per board had to stay low enough that spares were realistic.

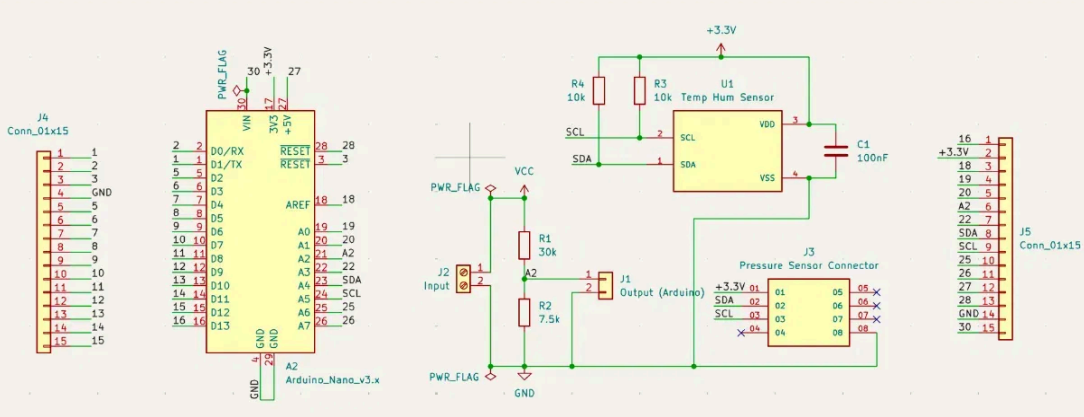
Secondary constraints shaped the layout. Every Arduino pin needed to stay accessible for debugging via DuPont jumpers, since the board would live inside a sealed enclosure and hands-on probing would be rare. The Arduino itself had to be socketed rather than soldered, so a failed Nano could be swapped without reflowing anything. And the mounting interface to Jill's board had to tolerate manufacturing variance on both sides, her board and mine are fabricated separately, so the connector geometry couldn't demand tight tolerances.

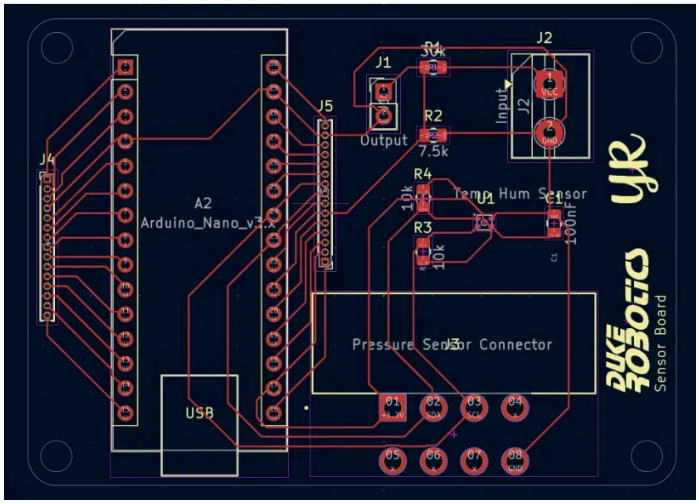
Design Iterations:

The board went through three major revisions before arriving at its final form. Each revision was driven by feedback from design reviews with Sagar, Sid, Patrick, and Jill, and each addressed a specific limitation of the previous version.

Version 1: Initial Design

The first version was a direct translation of the existing breadboard setup into PCB form. The Arduino Nano v3 sat at the center, with the SHT45 as a bare temperature and humidity IC wired directly to the Arduino's 3.3V I2C bus. The Bar30 pressure sensor connected through a DF13 header, also wired directly to the Arduino's I2C lines without any voltage translation. The voltage divider (R1 30kΩ, R2 7.5kΩ) scaled battery voltage down to A2 for ADC reading. The Arduino breakout used two 1×15 male pin headers (J4, J5), and the board had a 2-pin screw terminal input (J2) and 2-pin output header (J1). No interboard mounting existed yet, and the board was a standalone module.

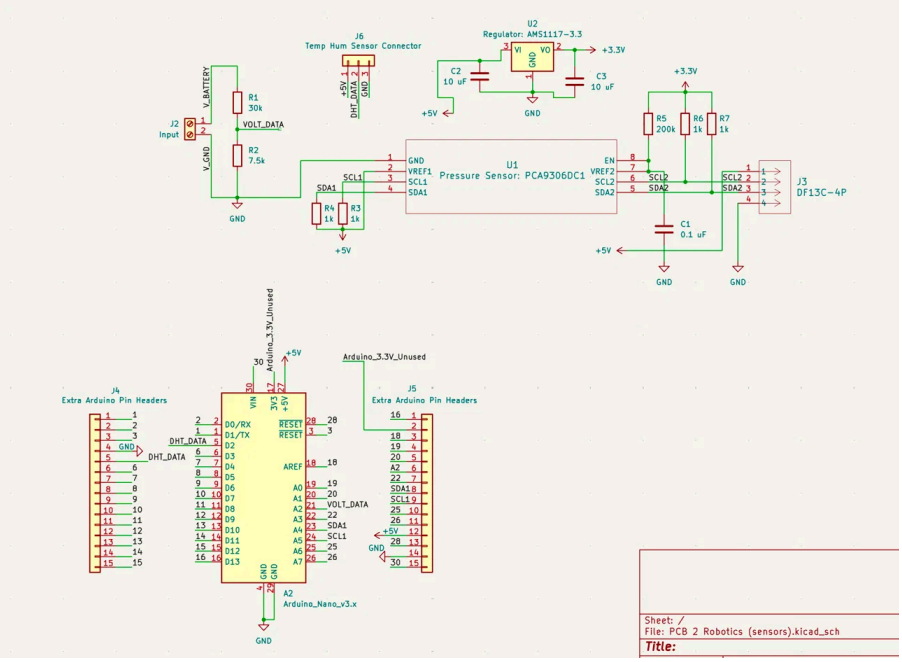




The main limitation surfaced in the first review: the SHT45 as a bare IC meant any sensor failure required reflow work, and the Arduino's 5V I2C bus was wired directly to the 3.3V Bar30, which worked but wasn't a safe long-term solution.

Version 2: First Review

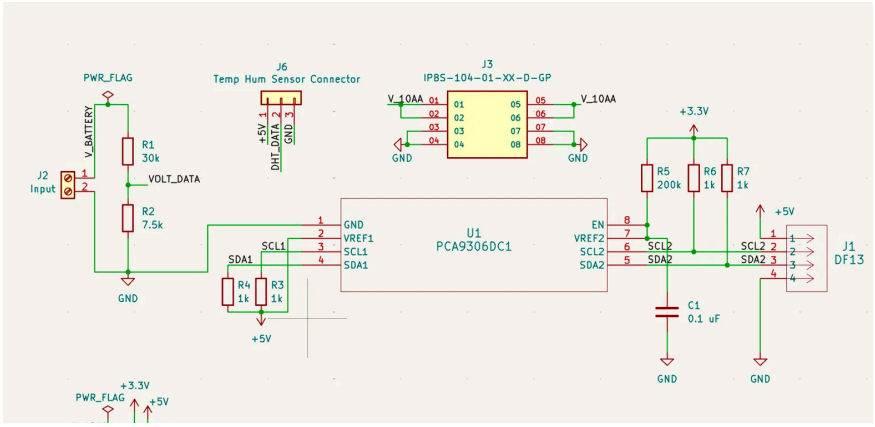
Three changes defined this revision. The SHT45 bare IC was removed and replaced with a 3-pin female header (J6) for the DHT11 breakout module, easier to swap, no reflow needed for sensor replacement. The PCA9306 I2C level shifter (U1) was added to properly bridge the Arduino's 5V I2C to the Bar30's 3.3V bus, with pull-ups on both sides, EN tied high, and VREF2 decoupled. And after Sid raised concerns about the Arduino's onboard 3.3V regulator being noisy and current-limited, an AMS1117 3.3V LDO (U2) was added with 10µF decoupling caps on input and output. The Arduino's 3.3V pin was disconnected from the rail and labeled "Arduino_3.3V_Unused."

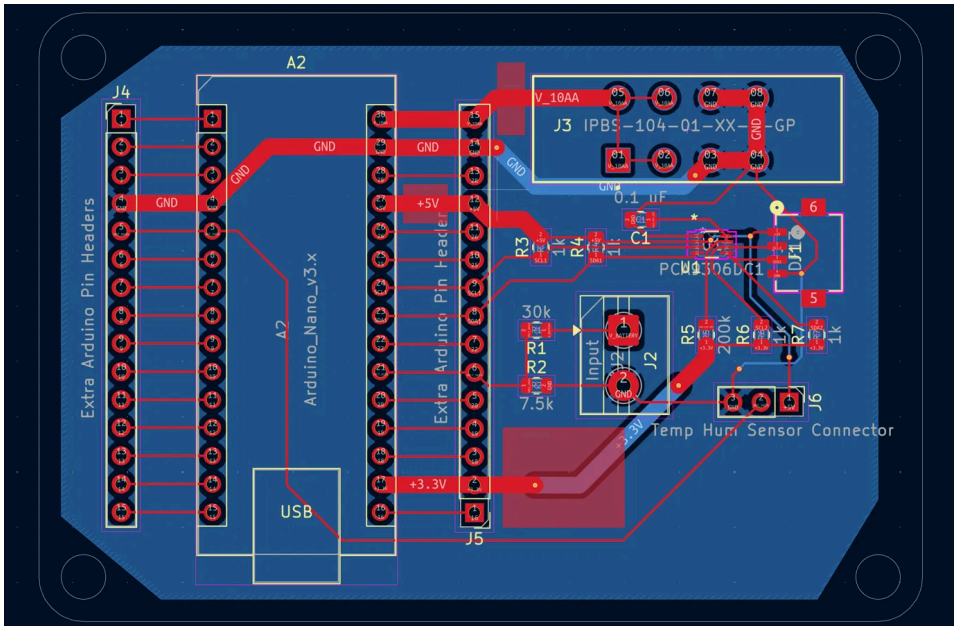


The sensor interfaces were now clean, but the LDO added parts count without a clearly justified benefit.

Version 3: Second Review Pass

Sagar and Patrick reviewed the LDO addition and concluded the Arduino's onboard regulator handled the actual current load fine. The AMS1117 was removed along with its decoupling caps, and the Arduino's 3.3V pin was reconnected to the +3.3V rail. In the same revision, Jill requested an interboard mounting solution so the sensor board could physically attach to her power board rather than exist as a loose module. A Samtec IPBS-104 dual-row connector (J3) was added to provide both mechanical attachment and electrical connection across the two boards.





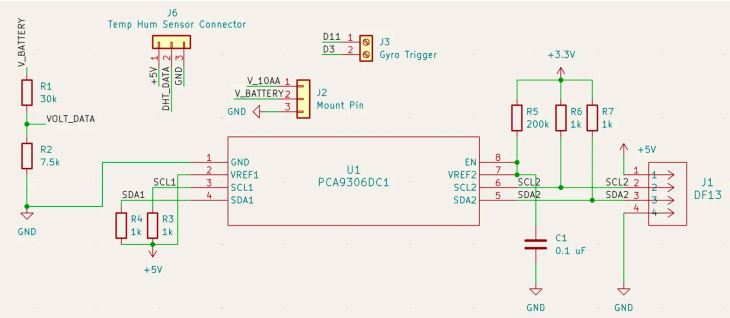
The IPBS-104 worked electrically but introduced two problems: it was oversized for the available board area, and sourcing a single connector from Samtec added cost and lead time compared to standard pin headers.

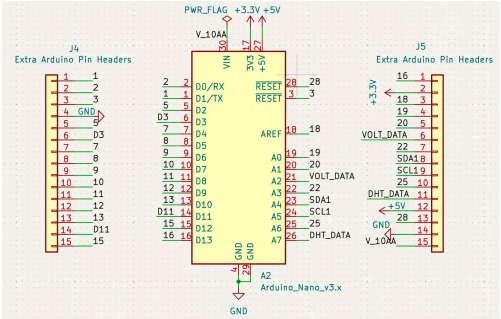
Version 4: Final Design

The final revision addressed the remaining mounting and layout concerns. The IPBS-104 was swapped for two 3-pin headers (J2 and J3), one on each side of the Arduino, giving better mechanical balance and using standard parts. Both mount headers were flipped to the back layer so the sensor board mounts on top of Jill's power board. The J4, J5, and J6 pin headers were converted from male to female sockets, allowing the Arduino and DHT11 module to plug in from above and be swapped without desoldering. A 2-pin screw terminal (J7) was added to expose Arduino pins D11 and D3 for the gyro trigger signal, per Sagar's final request. The board was compacted to 41.5 × 48 mm to fit the tight electrical stack. The full result is detailed in the Final Design section below.

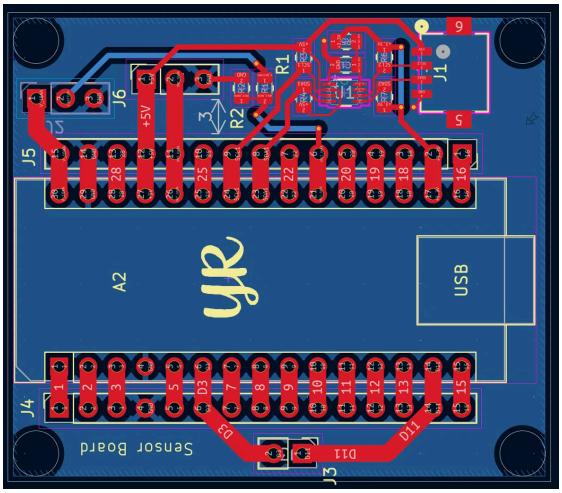
Final Design:

The board is organized around the Arduino Nano v3, with every sensor, connector, and support component wired through it. The design splits into three regions: Power and voltage sensing, I2C level-shifting cluster for the Bar30, Arduino breakout and interboard mounting pins.





- Power enters through J2, a 3-pin mount header on the bottom layer that connects to the power board. V_BATTERY is the raw battery voltage, used both to power the Arduino through its VIN pin and to be measured. The Nano regulates it internally to +5V and +3.3V for the rest of the board.
- Ground is a continuous copper pour on the back layer (shown in blue in the layout), providing low-impedance return paths and basic EMI shielding under the I2C traces. J3, also on the bottom layer, carries the gyro trigger signals (D11 and D3) across to the adjacent board.
- Battery voltage sensing uses a resistor divider formed by R1 (30 kΩ) and R2 (7.5 kΩ), which scales V_BATTERY down by a factor of 5 to keep it safely under the 3.3V ADC limit on pin A2. Firmware reads VOLT_DATA and converts the raw ADC value back to actual battery voltage.
- The I2C level shifter is the most involved subsystem. The Arduino's bus runs at 5V but the Bar30 requires 3.3V logic, so the PCA9306 (U1) bridges the two sides bidirectionally. Supporting components are R3 and R4 (1 kΩ pull-ups on the 5V side), R6 and R7 (1 kΩ pull-ups on the 3.3V side), R5 (200 kΩ pulling EN high to keep the shifter enabled), and C1 (0.1 µF decoupling the VREF2 reference). The shifted bus runs to J1, a DF13 4-pin connector matching Blue Robotics' standard Bar30 pinout: +5V, SDA2, SCL2, GND.
- The DHT11 temperature and humidity sensor plugs into J6, a 3-pin female socket wired to +5V, Arduino pin D2 (DHT_DATA), and GND. Socket format means the breakout module can be swapped without reflowing anything.



- Every Arduino pin is broken out to two 1×15 female sockets, J4 and J5, running along the sides of the board. These give full access to every Nano pin via DuPont jumpers for debugging or future expansion, and since the Arduino itself drops into these same sockets, the Nano can be swapped without touching any solder joints.
- Four M3 mounting holes (3.4 mm diameter) sit at the corners for fastening to Jill's board below.

Manufacturing:

The board was fabricated through JLCPCB in a batch of 5. Specifications: 2-layer FR-4 (TG135) with 1.6 mm thickness, 1 oz outer copper, blue solder mask, white silkscreen, and lead-free HASL surface finish. Minimum via hole size was 0.3 mm, and tented vias were specified for cleaner silkscreen coverage. No assembly or stencil was ordered since the components would be hand-soldered at the Duke Robotics workspace.

Gerber files were exported from KiCad with the following layers included: F.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, F.Fab, B.Fab, and Edge.Cuts. Drill files were exported in Excellon format with millimeter units and absolute coordinates, combining plated and non-plated holes into a single file. Coordinate format was set to 4.6 mm with X2 extended Gerber attributes enabled, which gives JLCPCB's parser full netlist information for automated electrical verification. The full export was zipped and uploaded through JLCPCB's web interface, with the board outline confirmed at 41.5 × 48 mm during their file review.

Fabrication took approximately 3 days plus shipping. Boards arrived in early November 2026, weighing roughly 32 g each. Visual inspection on arrival confirmed clean solder mask coverage with no exposed copper where it wasn't intended, crisp white silkscreen across all component labels and the YR signature, and all four M3 mounting holes drilled to the specified 3.4 mm diameter. No visible fabrication defects were present on any of the 5 boards.

Bill of Materials:

BThe final BOM is 12 line items across passive components, one IC, the Arduino Nano, connectors, and headers. Total cost comes to roughly $69 for all packs, or about $5 per PCB when amortized across the component quantities ordered. The two most expensive line items per board are the Arduino Nano v3 (~$1.80) and the DF13 connector for the Bar30 (~$1.12), together accounting for over half the per-board cost. Passive components (resistors and the 0.1 µF capacitor) were sourced through Amazon assortment packs from Chanzon, while the PCA9306 level shifter, the DF13 connector, and the precision headers were ordered individually from Digikey. The Arduino Nano was already in Duke Robotics club stock.

ID	Title	Footprint	Qt.	Designation	Supplier	Pack	Per PCB
1	R2	R_0805_2012Metric	1	7.5 kΩ	Amazon	4.99	0.0499
2	R5	R_0805_2012Metric	1	200 kΩ	Amazon	4.99	0.0499
3	R3,4,6,7	R_0805_2012Metric	4	1 kΩ	Amazon	5.99	0.2396
4	J4, J5	PinSocket_1x15_P2.54 mm_Vertical	2	Pin Sockets	Amazon	8.69	0.5793

5	C1	C_0805_2012Metric	1	0.1 μF	Amazon	8.88	0.0444
6	U1	SOT765-1_NXP	1	PCA9306DC1	Amazon	17.99	1.799
7	J3	PinHeader_1x02_P2.54mm_Vertical	1	Gyro Trigger	Amazon	1.51	0.0151
8	J6	PinSocket_1x03_P2.54mm_Vertical	1	Temp Hum	Digikey	5.6	1.12
9	A2	Arduino_Nano	1	Arduino	Club		
10	R1	R_0805_2012Metric	1	30 kΩ	Amazon	4.99	0.0499
11	J1	CON4_1X4_U_DF13	1	DF13	Digikey	3.65	0.73
12	J2	PinHeader_1x03_P2.54mm_Vertical	1	Mount Pin	Digikey	1.5	0.3
Total Cost | Total Cost per PCB						$68.78	$4.98

Challenges and Lessons:

The first real challenge was KiCad library management. The initial setup had component libraries scattered across download folders named by date, which caused path warnings every time the project opened and symbol mismatches whenever a footprint got reassigned. The fix was mechanical but worth the time: consolidate everything into a single stable directory with descriptive folder names, then re-register each library through KiCad's preferences. The cleanup took an afternoon but eliminated an entire category of errors that would have resurfaced every revision.

The second was ERC pedantry. KiCad's Electrical Rules Checker treats connector pins as "passive" by default, which means power that passes through a connector to reach the rest of the board gets flagged as "input power pin not driven by any output." The warnings look alarming but are almost always cosmetic — the circuit is electrically correct, KiCad just can't trace power through a passive symbol. The lesson was learning to distinguish genuine electrical issues from these false warnings, and documenting each exclusion with a reason so future-me (or anyone else opening the project) knew why it was ignored.

The third was smaller but surprisingly stubborn: resizing the YR signature footprint. KiCad's Footprint Editor doesn't have a built-in scale tool for imported graphics, so shrinking the signature required re-importing the original image through the Image Converter at a smaller target size and overwriting the old footprint file. Not hard once figured out, but it took longer than it should have.

The biggest process lesson was the value of mid-process design reviews. Catching the SHT45-to-DHT11 swap, the LDO debate, and the mounting connector change before submitting the Gerbers meant none of those decisions required a costly board respin. Every revision caught something that would have otherwise shown up as an unusable board or a last-minute wiring hack. The time cost of running the design past Sagar, Sid, Patrick, and Jill was small compared to the cost of being wrong in silicon.

Future Plans:

Components are arriving within the next 1-2 weeks, after which assembly begins. The soldering order is SMD first, the PCA9306 and 0805 passives, followed by through-hole components (headers, DF13 connector, screw terminal), with the Arduino Nano installed last through its female sockets so it can be swapped without desoldering. Hand-soldering at the Duke Robotics workspace, using flux paste and a fine-tip iron for the PCA9306's VSSOP-8 package.

Testing follows a fixed sequence. First, continuity and short checks with a multimeter across the power rails before applying any voltage. Then a power-on test with current monitoring to catch any unexpected draws. Once stable, an Arduino sketch runs an I2C address scan to confirm the Bar30 is detected at its expected address, followed by a voltage divider calibration — measuring actual battery voltage with a multimeter and comparing it to the ADC reading to derive the scaling constant for firmware. Finally, individual sensor read-outs printed to Serial to verify the DHT11 and Bar30 return sensible values.

Rev 2 ideas are already queued: a heartbeat LED that Sid originally suggested, a reset header for easier Arduino debugging in-situ, and potentially consolidating this board with the thruster MCU if the ongoing RP2040 vs STM32 evaluation produces a clear winner. Pads for a future IMU integration would also be worth adding if the gyro screw terminal proves limiting.

Longer-term, the final design, schematic, and PCB files will be committed to the Duke Robotics repo as a reusable module so future robots can start from a known-good baseline rather than re-deriving the wiring from scratch. Both Oogway and Crush will run this board as their standard sensor interface going forward.

Acknowledgements:

- Yuvraj Ruparel: Main designer of PCB
- Saagar Arya: Design Reviewer, Ex-President
- Jill Wang: PCB coordinations, Electrical Lead
- Sid Kini: Schematic review, Current President
- Patrick Zheng: Schematic Reviewer, Ex-President